Attributing Inference-Time Gains in Lean Theorem Proving

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METHODS • CONTROLLED STUDIES • VERIFIABLE EVIDENCE

A fixed-budget portfolio is a controlled probe: separate search policy, retrieval, realized compute, and prover dependence before assigning credit to an improvement.

Host: DeepSeek-Math-7B-Base + LoRA • 60 validation / 60 test theorems • theorem-level wall-clock accounting

01 From proof-state search to an attributable intervention

A. Host and optional retrieval

Target + Lean environment
goal state Γ |- q



Mathlib dependency graph
150,161 nodes
256 Laplacian coordinates
optional premise retrieval

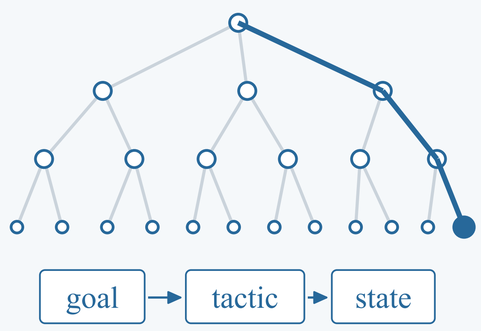
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Proof-state backbone
DeepSeek-Math-7B + LoRA

Retrieval is an experimental factor. The no-retrieval arm tests whether scheduling can help on its own.

B. Reallocate one budget across heterogeneous attempts

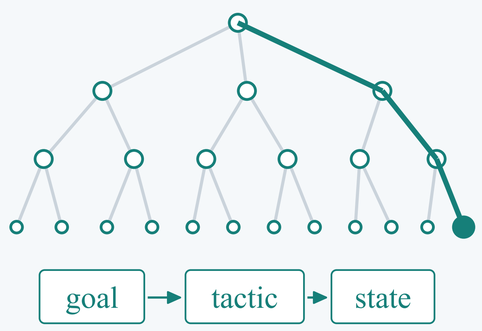
Base expansion



ordinary search profile

Spend the initial slice exploring a compact tactic frontier. Stop when a valid proof is found.

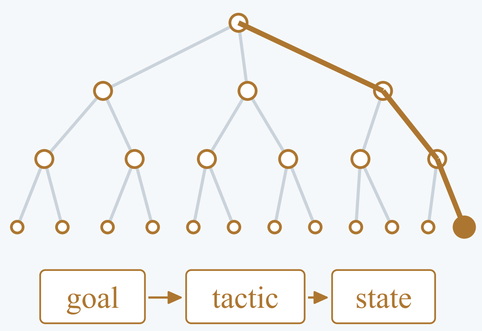
Broader frontier



alternative expansion pattern

Use a different search profile to diversify states reached within the same theorem run.

Heavy fallback



continue unresolved targets

Allocate the remaining opportunity to difficult targets; preserve successful earlier attempts.

Single / Balanced / Two-stage / Three-stage
One theorem run; internal attempts do not become Pass@3.

C. Time ledger

init / actor | attempt 1 | attempt 2 | fallback

Σ attempt time + overhead ≤ B

Budget utilization is measured on complete theorem runs, including environment setup and orchestration.

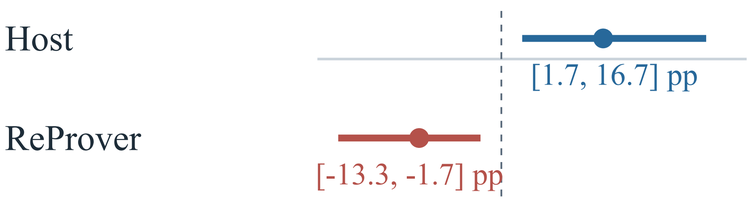
Schematic trees and allocations; only charts and tables below encode measured observations.

02 Effect and portability

No retrieval	Single	Portfolio	Δ pp
Host valid.	28.3	36.7	**+8.3**
Host test	31.7	35.0	**+3.3**
ReProver valid.	45.0	38.3	**−6.7**
ReProver test	40.0	38.3	**−1.7**

Pass@1 (%) • 60 targets per split

Paired validation intervals



Paired bootstrap 95% confidence intervals

The host-local effect survives retrieval removal. Its direction reverses in generator-only ReProver, including clean reruns.

Test Δ intervals: host [0.0, 8.3] pp; ReProver [−5.0, 0.0] pp.

03 Budget use explains part

Runs using ≥55 s of nominal 60 s



43.3% of single-profile runs stop within 15 seconds. Portfolio runs reach the near-budget region much more often.

Matched-compute follow-up

15% → 30%

A stronger single profile recovers substantial performance on the 20-theorem control subset. Equal nominal limits do not imply equal realized computation.

The intervention is best interpreted as a budget-allocation policy that invests more in unresolved hard cases.

04 Component attribution

Retrieval-enabled host comparison

Split	Single	Portfolio
Validation	28.3%	**31.7%**
Test	26.7%	**31.7%**

Positive scheduling effects also appear with retrieval. Strict-core controls do not establish the spectral retriever as a strong component; some matched-budget retrieval effects are negative.

A control matrix for the claim

Remove retrieval
Separate scheduling from premise access.

Match actual compute
Challenge an underused single baseline.

Change prover
Test sensitivity to the host implementation.

05 Theorem-level evidence and experimental design

Comparison	Retained	New solves	Lost solves
Host validation	17 / 17	5	**0**
Host test	19 / 19	2	**0**
ReProver validation	23 / 27	0	**4**

A theorem counts as solved when any internal attempt produces a Lean-verified proof within its total budget. The 60/60 split is complemented by seed-42 subsets of 20 for repeated controls and budget sweeps.

06 Host setup and scope of the conclusion

261,393 training proof states from 51,406 theorems; 14,031 validation states from 2,706 theorems. Splits are disjoint by theorem identity. LoRA: rank 16, α = 32, dropout 0.05.

Giant component: 148,971 nodes; 119/120 targets covered.
Attempt traces, full wall-clock accounting, clean reruns, and error accounting tie the measured effect to the executed policy. The evidence supports a local gain and a portability boundary, rather than a universal portfolio advantage.